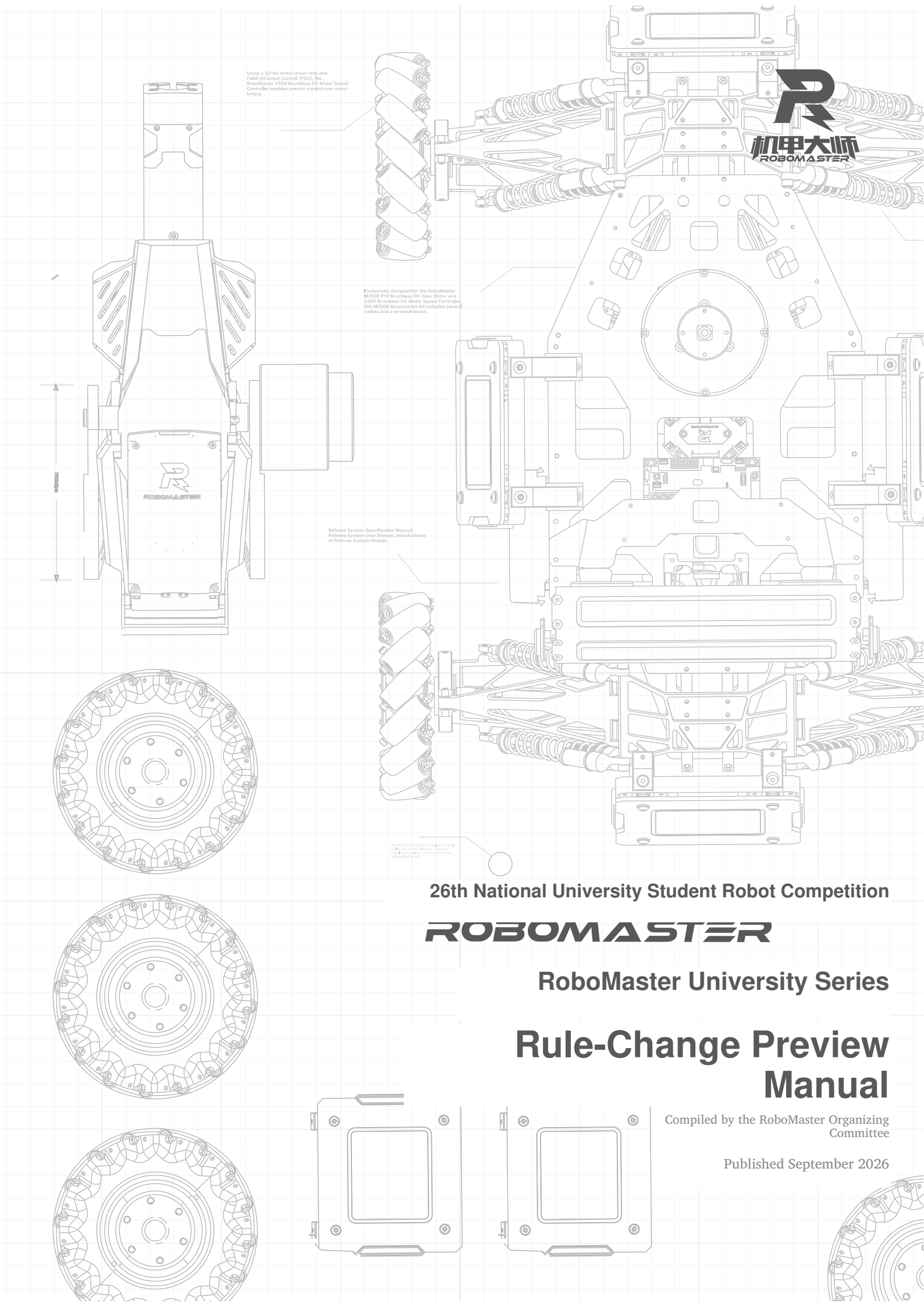


26th National University Student Robot Competition

ROBOMASTER

RoboMaster University Series

Rule-Change Preview Manual

Compiled by the RoboMaster Organizing Committee

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1 Overview of Main Changes in the New Season

This chapter briefly describes the main rule adjustments for the RM2027 season. Compared with RM2026, the main revisions are as follows:

Robot systems:

- Redesigned the Aerial Robot's deployment, use, and functions.
- The former Hero Robot and Engineer Robot are merged into the Heavy Robot.
- The former Custom Controller and Custom Client are merged into the Custom Terminal.

Battlefield:

- The former Launch Ramp is replaced by a Tunnel.
- Adjusted the overall layout of the Operator Room.
- Added the High Point, Hangar Placement Zone, and Workbench.
- Removed the Flight Zone and Wireless Charging Device Placement Zone.



This manual presents only some key rule changes. For complete rule details and final enforcement standards, refer to the official Rules Manual to be published later.

2 Robot System Mechanics and Building Specifications

2.1 Robots and Operators

The robot lineup is as follows:

Table 2-1 Robot Lineup

Type	Full lineup quantity (units)	Competition stage
Heavy Robot	1	Throughout RMUC 2027
Infantry Robot	2	
Aerial Robot and Hangar	1	
Sentry Robot	1	
Dart System	1	
Radar	1	

Minimum lineup for the first round of each match: at least 3 robots, excluding the Radar and Dart System.

The Operator lineup is as follows:

Table 2-2 Operator Lineup

Type	Robots operated	Full lineup headcount
Ground Robot Operator	Heavy Robot	1
	Infantry Robot	2
	Sentry Robot	0
Commander	Aerial Robot and Hangar (optional), Dart System (optional)	1

2.1.1 Heavy Robot

The Heavy Robot can gain diverse team Buffs by acquiring and assembling Energy Units. Once certain conditions are met, it can also launch 42mm projectiles.

Table 2-3 Heavy Robot Key Features

Key feature	Description
Initial zone	Starting Zone
Operation mode	Unrestricted; at most 1 Remote Controller and 1 Custom Terminal may be configured
Multi-robot communication	Allowed
Launching mechanism	1 installable modular 42mm launching mechanism
Projectile loading and resupply	- During the three-minute Setup Period, 42mm and 17mm projectiles can be preloaded in both the robot and the modular 42mm launching mechanism; the initial Projectile Allowance is 0. • During the seven-minute Round, Projectile Allowance can be exchanged at permitted Buff Points or exchanged remotely.
Experience and Performance System	Not applicable Note: The attributes of the Heavy Robot change as the match progresses.

2.1.1.1 Special Mechanics

The relationship between the Heavy Robot's attributes and its highest completed Assembly Level is shown in the table below:

Table 2-4 Heavy Robot Assembly Level Attributes

Highest completed Assembly Level	Max HP	Barrel Heat Limit	Barrel Cooling Rate	Initial Launch Speed limit
Start of match	400	-	-	-
Level 3	600	300	100	16.5m/s
Level 4	600	300	100	16.5m/s

Modular 42mm launching mechanism and unlocking:

At the start of a match, the Heavy Robot's modular 42mm launching mechanism is stored on a Workbench in the Resource Zone and is not installed on the robot. Once any one of the following conditions is met, the Heavy Robot unlocks permission to use it, may install it on the robot, and may launch 42mm projectiles.

The modular launching mechanism unlocks when any of the following conditions is met:

- Condition 1: Complete Level 3 assembly.
- Condition 2: 5 minutes of the match have elapsed.
- Condition 3: For every specified amount by which the team falls behind, the unlock time in Condition 2 is brought forward.

In RMUL, the Heavy Robot has no assembly tasks and unlocks the modular 42mm launching mechanism at the start. It must still install it on the robot during the match before it can launch 42mm projectiles. The exact mechanics and values will be announced in the final version.

2.1.1.2 Building Specifications

The Heavy Robot must meet the following requirements:

- The Heavy Robot must be able to move horizontally and pick up Energy Units in the Resource Zone.

  In RMUL, the Heavy Robot does not need to be able to pick up Energy Units.

- Heavy Robot building parameters:

Table 2-5 Heavy Robot Building Parameters

Item	Limit	Remarks
Operation mode	Unrestricted; at most 1 Remote Controller and 1 Custom Terminal may be configured	-
Max total power supply capacity (Wh)	300	-
Max operating voltage (V)	30	The voltage at any point inside the robot at any time relative to the battery negative terminal must not exceed 30V
Launching mechanism	Modular 42mm launching mechanism	Can only be installed during the seven-minute Round
Max weight	• 36: Includes the battery and Referee System weight, excluding	-

Item	Limit	Remarks
	Weight of the modular 42mm launching mechanism. 41: Includes the battery, Referee System, and modular 42mm launching mechanism.	
Maximum stowed dimensions (mm, L × W × H)	The robot has a state with length, width and height not greater than 600 × 600 × 600	-
Maximum extended dimensions (mm, L × W × H)	One of the following requirements must be met: Requirement 1: Under all circumstances, the robot's length, width, and height must not exceed 1200 × 1200 × 1100. Requirement 2: To be specified in the final rules.	-
Referee System	- Small Armor Module • Video Transmitter Module with camera (transmitter end) • Field Interaction Module • Positioning Module (not applicable in RMUL) • Main Controller Module • Power Management Module • Light Indicator Module	-

The modular 42mm launching mechanism must meet the following requirements:

- The Heavy Robot's main body must be electrically isolated from the modular 42mm launching mechanism. The working voltage inside the Heavy Robot main body must not exceed 30 V or interfere with normal operation of the Referee System.
- The modular 42mm launching mechanism must be capable of launching 42mm projectiles.
- Modular 42mm launching mechanism building parameters:

Table 2-6 Modular 42mm Launching Mechanism Building Parameters

Item	Limit	Remarks
Max total power supply capacity	-	No battery may be installed; powered by the robot

Item	Limit	Remarks
		Direct power supply
Maximum working voltage (V)	140	The voltage at any point inside the module at any moment must not exceed 140V relative to the battery negative terminal
Maximum weight (kg)	6	Includes Referee System weight
Maximum stowed dimensions (mm, L × W × H)	The modular 42mm launching mechanism has a state with length, width and height not greater than 400 × 400 × 400	-
Referee System	42mm Speed Monitor Module	

The Heavy Robot may perform the following operations, but only at the Workbench:

Disassemble into sub-robots or multiple subsystems connected by flexible cables, or reassemble them from these states.

Install or remove the modular 42mm launching mechanism.

2.1.2 Aerial Robot and Hangar

The Aerial Robot may launch from the Hangar and perform diverse attack and reconnaissance tasks on the battlefield.

Table 2-7 Aerial Robot and Hangar Key Features

Key feature	Description
Initial zone	Hangar Placement Zone
Multi-robot communication	Allowed
Launching mechanism	Not applicable
HP Recovery and Revival mechanics	Not applicable
Experience and Performance System	Not applicable

Typical Aerial Robot functions:

- Damage the opposing Outpost and Base Armor Modules by direct impact.
- Intercept by actively colliding with and changing the flight paths of opposing Darts and projectiles.
- Hover to gather information, then return to the Hangar and transmit it for further use by systems such as the Radar.

2.1.2.1 Building Specifications

The Aerial Robot must meet the following requirements:

- The Aerial Robot must be capable of hovering continuously for at least 10 seconds.
- A fully enclosing propeller guard must be installed around the Aerial Robot. The propellers and body must not be exposed. When the Aerial Robot collides with an Armor Module at any angle and a certain horizontal speed, the guard must effectively prevent the propellers from contacting the Armor Module.
- The Aerial Robot's propellers must be made of flexible material.
- The Aerial Robot must have a Remote Controller/receiver for emergency propeller shutdown and must have a safety officer.
- Aerial Robot building parameters:

Table 2-8 Aerial Robot Building Parameters

Item	Limit	Remarks
Operation mode	Fully automatic	Can communicate with the Hangar while inside the Hangar
Max total power supply capacity (Wh)	TBD	-
Max operating voltage (V)	TBD	-
Launching mechanism	Installation of a launching mechanism is not allowed	-
Max weight (kg)	y	Size and weight limits are linearly related: $y=-0.001225x+0.3225$, $y\in[0.15,0.249]$
Maximum extended dimensions (mm)	When the Aerial Robot weighs y, its length, width, and height must not exceed x × x × x under any circumstances.	

The Hangar must meet the following requirements:

- The Hangar must be capable of storing the Aerial Robot.

- Hangar building parameters:

Table 2-9 Hangar Building Parameters

Item	Limit	Remarks
Operation mode	Unrestricted; at most 1 Remote Controller may be configured for debugging	-
Max total power supply capacity (Wh)	300	-
Max operating voltage (V)	30	The voltage at any point inside the Hangar at any time relative to the battery negative terminal must not exceed 30V
Aerial Robot load limit	4	-
Max weight (kg)	TBD	-
Max stowed dimensions (mm, L × W × H)	300*300*200	-
Max extended dimensions (mm, L × W × H)	TBD	-
Referee System	TBD	-

The Hangar may provide initial propulsion to the Aerial Robot, but the direction must be within a 15° cone around the Hangar's vertical normal.

2.2 Custom Terminal

The Custom Terminal (formerly the Custom Controller and Custom Client) is a multipurpose device built by the team. It monitors robot motion and status, controls robot actions, receives match-related information and robot video from the Referee System Game Engine Server, and can send competition-specific interactive commands to that server to expand human-machine interaction.

A typical Custom Terminal may consist of a display, a computing platform (including memory, controller, and processor), and input devices. It may also be integrated into one device, such as a handheld tablet with an Ethernet port.

Example: A team builds a Custom Terminal that receives robot video and displays it on a custom screen with a custom UI overlaid through a nonofficial link, while a paired controller operates the robot. The team can therefore complete a match without using the official Referee System Operator End or a Remote Controller.

All Custom Terminals in the Operator Room may be connected to and communicate with one another by wire. A Custom Terminal may also connect to the Game Engine Server through the RJ45 port provided in the Operator Room and send and receive related content.

A Custom Terminal may receive match information displayed on the Referee System Operator End and available to its team, such as both teams' HP, its team's economy, remaining match time, and announcements of key match events.

A Custom Terminal may send the Referee System match commands supported by the Operator End, such as selecting a Performance System before the match, exchanging for permitted Projectile Allowance during a match, and intervening in Sentry actions.

2.2.1 Building Specifications

The Custom Terminal must meet the following building parameters:

Table 2-10 Custom Terminal Building Parameters

Item	Limit	Remarks
Max operating voltage (V)	30	The voltage at any point inside the module at any time relative to the battery negative terminal must not exceed 30V
Max stowed dimensions (mm, L × W × H)	600*600*600	-
Max extended dimensions (mm, L × W × H)	Unlimited	Consequences caused by size, such as inconvenient use or inability to compete normally, are borne by the participating team
Data transmission	Video feed information: UDP Match-related data: MQTT + ProtoBuf	-
Protocol	See the Communication Protocol	-

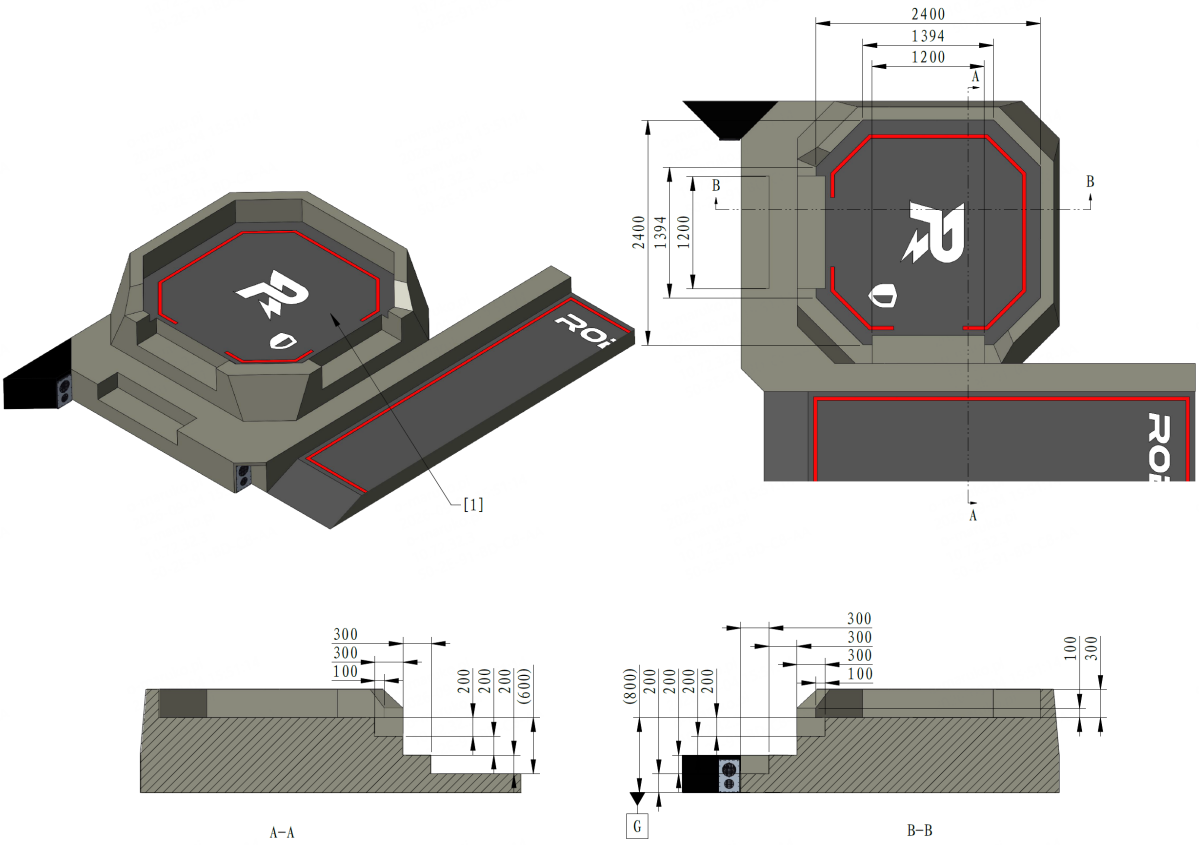
Usage requirements:

The Custom Terminal must not use wireless transmitters or receivers.

3 Battlefield and Related Mechanics

3.1 High Point

The High Point is in the Base Zone. Three steps connect the trapezoidal high ground to the High Point, and robots may climb them to enter it.



[1] High Point

Figure 3-1 High Point Diagram

3.1.1 Buff Point Mechanics

All allied robots may occupy their team's High Point Buff Point.

If a team's High Point Buff Point is occupied by one of its robots, that robot receives a 50% defense Buff and may exchange non-remotely for permitted Projectile Allowance.

3.2 Tunnel

The former Launch Ramp is replaced by a Tunnel in the Highway Zone. Robots may pass through or over the Tunnel to reach the opposing half of the battlefield quickly.

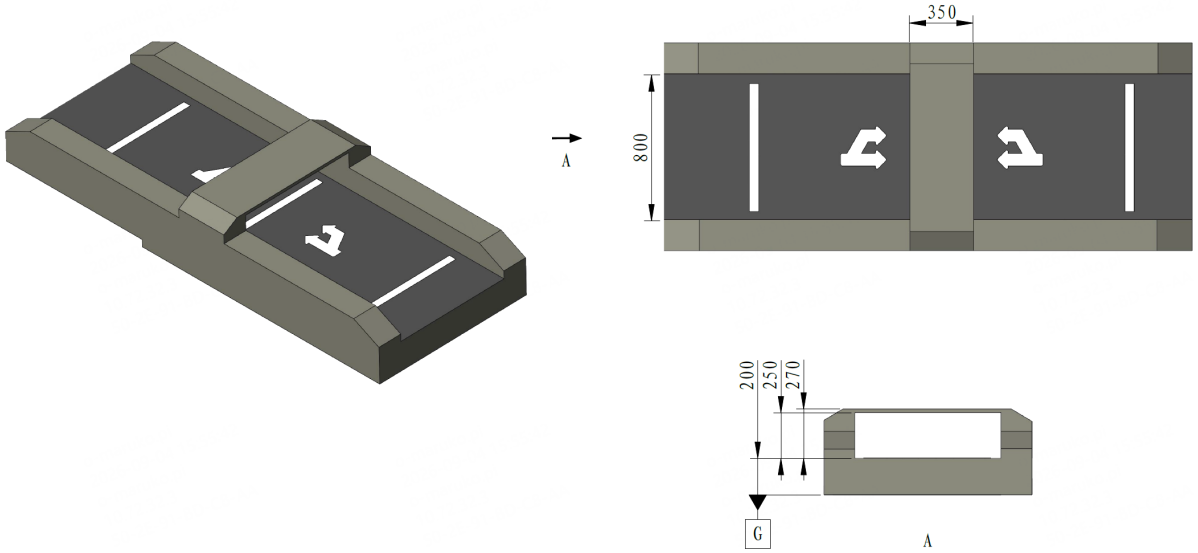


Figure 3-2 Tunnel Diagram

3.2.1 Buff Point Mechanics

The two types of buffs differ according to the type and sequence of Field Interaction Module cards detected while passing through the Tunnel:

Table 3-1 Tunnel Buff Trigger Logic and Effects

Buff type	Field Interaction Module card types and order detected	Buff effect
Leap over the Tunnel	Only the Field Interaction Module cards at the Tunnel entrance and exit are detected, in order	25% defense Buff, lasting 30 s
Pass through the Tunnel	The Field Interaction Module cards at the Tunnel entrance, interior and exit are detected, in order	1.5x Barrel Cooling Rate Buff, lasting 120 s

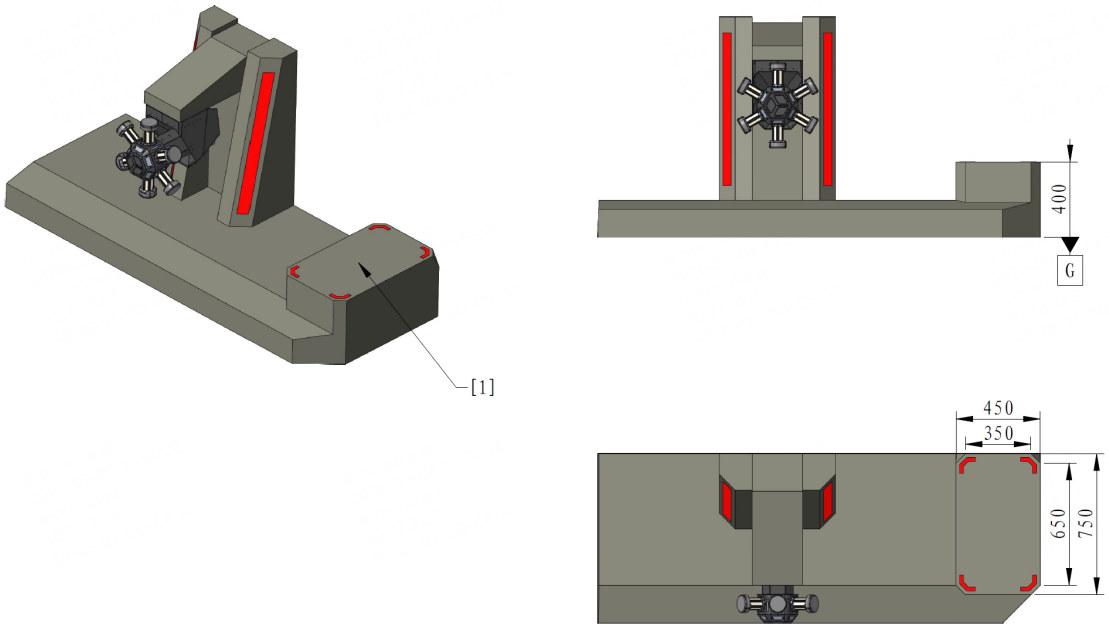
A robot must detect each Field Interaction Module card in the Tunnel in sequence within 3 seconds, without detecting any other type of Field Interaction Module card along the way, to obtain the above Buff.

3.3 Supply Zone

The Supply Zone includes the Resource Zone, Supply Zone Buff Point, and Workbench. It is an important area for robots to restore HP, replenish Projectile Allowance, obtain Energy Units, and assemble the modular 42mm launching mechanism.

3.3.1 Workbench

The Workbench is in the Supply Zone. Robots may pick up and assemble the modular 42mm launching mechanism there.

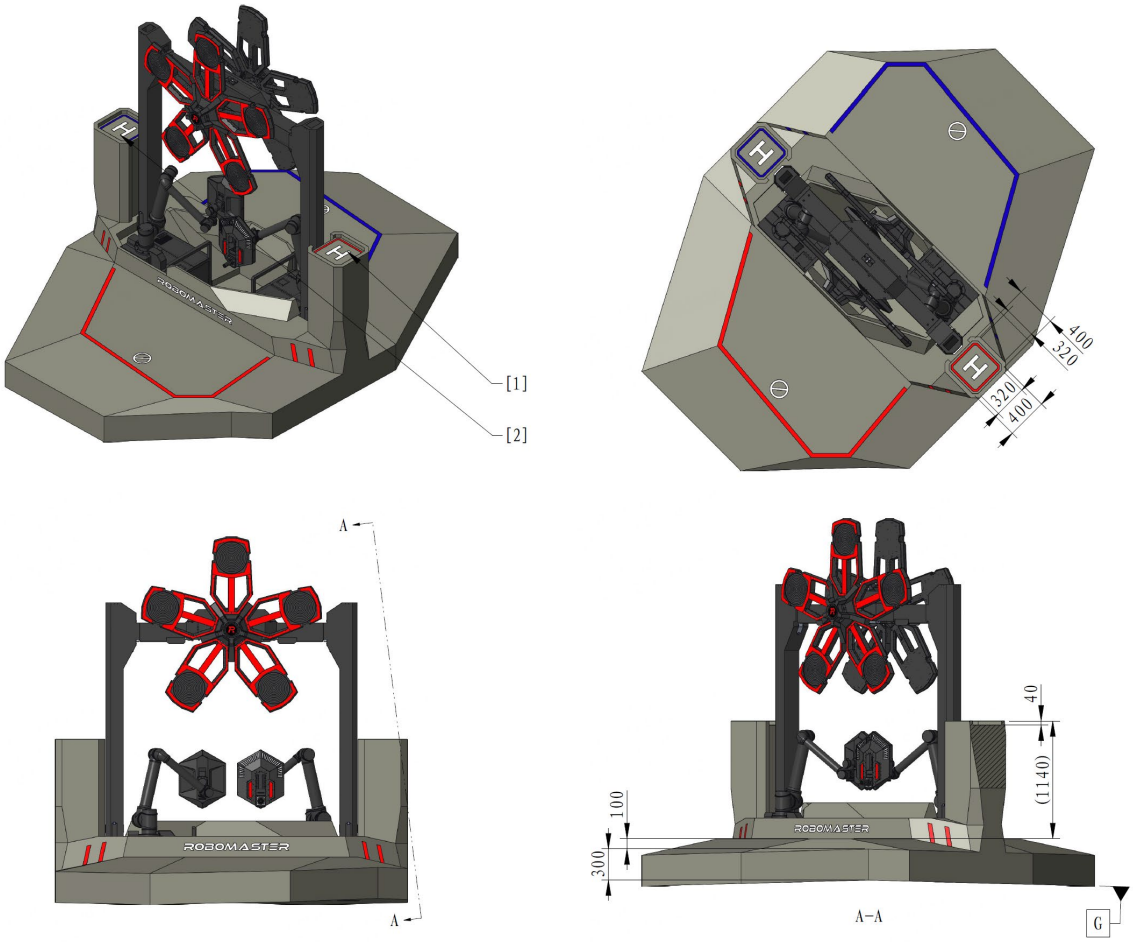


[1] Workbench

Figure 3-3 Workbench Diagram

3.4 Hangar Placement Zone

The Hangar Placement Zone is in the center of the battlefield and is divided between the Red and Blue sides. Each side may install its Hangar only in its own Hangar Placement Zone.



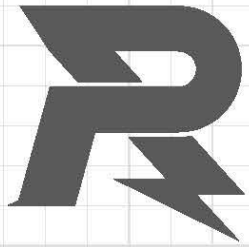
[1] Red Hangar Placement Zone

[2] Blue Hangar Placement Zone

Figure 3-4 Hangar Placement Zone Diagram

3.5 Operator Room

The Operator Room is near the perimeter of the battlefield and is where Operators stay during matches. Each Operator desk is 120 cm long and 75 cm wide and is equipped with the corresponding number of computers. Each computer has a monitor, mouse, keyboard, Camera Video Transmitter Module (Receiver) and its stand, one 1GbE (RJ45) port, wired headphones, and other official equipment.



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